

```

import sqlite3

import pandas as pd

from collections import defaultdict


# =====

# CONFIGURATION

# =====

DB_PATH = "database.sqlite"


SHEET_0 = "data/shipment_info.csv"
SHEET_1 = "data/shipment_products.csv"
SHEET_2 = "data/shipment_routes.csv"


# =====

# DATABASE CONNECTION

# =====

conn = sqlite3.connect(DB_PATH)

cursor = conn.cursor()


# =====

# PART 1: LOAD SPREADSHEET 0 (DIRECT INSERT)

# =====

def load_sheet_0():

    df = pd.read_csv(SHEET_0)

    for _, row in df.iterrows():

        cursor.execute("""

```

```
INSERT INTO shipments_sheet0 (shipment_id, product, quantity)
VALUES (?, ?, ?)

""" , (
    row["shipment_id"],
    row["product"],
    row["quantity"]
))
```

```
conn.commit()
print("Sheet 0 inserted successfully.")
```

```
# =====
# PART 2: PROCESS SPREADSHEET 2 (ROUTES)
# =====

def load_routes():
    df = pd.read_csv(SHEET_2)

    routes = {}

    for _, row in df.iterrows():
        routes[row["shipment_id"]] = {
            "origin": row["origin"],
            "destination": row["destination"]
        }

    return routes
```

```

# =====

# PART 3: PROCESS SPREADSHEET 1 (GROUP BY SHIPMENT)

# =====

def load_sheet_1(routes):
    df = pd.read_csv(SHEET_1)

    # Group products by shipment_id
    shipment_map = defaultdict(list)

    for _, row in df.iterrows():
        shipment_id = row["shipment_id"]
        product = row["product"]
        quantity = row["quantity"]

        shipment_map[shipment_id].append((product, quantity))

    # Insert into database
    for shipment_id, products in shipment_map.items():

        if shipment_id not in routes:
            continue

        origin = routes[shipment_id]["origin"]
        destination = routes[shipment_id]["destination"]

        for product, quantity in products:
            cursor.execute("""
                INSERT INTO shipments_sheet1 (
                    shipment_id,

```

```
        origin,  
        destination,  
        product,  
        quantity  
    )  
VALUES (?, ?, ?, ?, ?)  
""", (  
    shipment_id,  
    origin,  
    destination,  
    product,  
    quantity  
))
```

```
conn.commit()  
print("Sheet 1 inserted successfully.")
```

```
# =====
```

```
# MAIN EXECUTION
```

```
# =====
```

```
def main():
```

```
    print("Starting ETL process...")
```

```
    load_sheet_0()
```

```
    routes = load_routes()
```

```
    load_sheet_1(routes)
```

```
print("ETL process completed successfully.")
```

```
conn.close()
```

```
if __name__ == "__main__":
```

```
    main()
```